

3D Printed Aircraft at Virginia Tech

Vortex

2026 Competition Report



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Table Of Contents

1. Team Structure	3
2. Aircraft Design	4
2.1 Configuration	4
2.2 Aerodynamics	6
2.3 Propulsion and power	7
2.4 Systems and electronics	8
2.5 Structures	9
3. Print and Manufacturing	12
4. Innovation, Lessons	19

2. Aircraft Design

2.1 Configuration

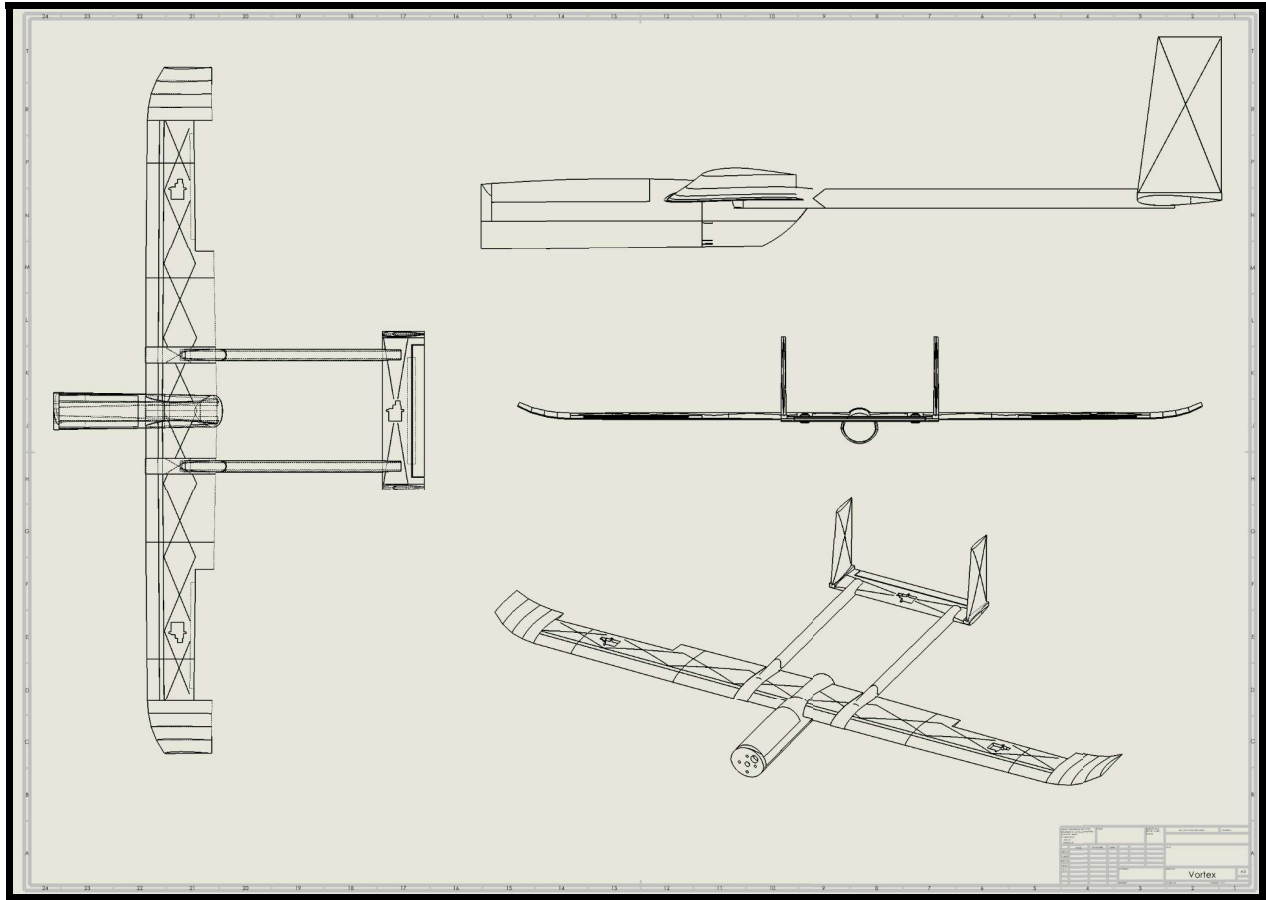


Figure 1. Vortex complete assembly engineering drawings(Front, side,top and isometric views)

3DPAC Team 2 at Virginia Tech has designed an aircraft known as “Vortex”. The aircraft follows a pod and boom configuration. A pod and boom aircraft is characterised by a small, central fuselage attached to the wing. The main caveat of this design is that the fuselage does not extend to the tail as seen in traditional aircraft designs. Instead, the two booms which originate from the wings extend to the tail.

This configuration was of considerable interest to the team due to the potential benefit of positioning the center of gravity close to the wing’s aerodynamic center (near the quarter chord point). In an unpowered glide, this position is favourable as it minimizes the nose down pitching moment, improving maneuverability which is especially important when considering the confined space the aircraft has to glide.

In addition to these benefits, this design distributes the load from the tail across two booms. By consequence, this configuration improves structural rigidity and can also accommodate a larger horizontal stabilizer without the added weight of an extended fuselage

required to support it. For a low-speed glider, a larger horizontal stabilizer improves elevator authority and pitch control, all in all allowing for a more precise glide and potentially increasing glide time. An initial design for the wings and tail was created in XFLR5, as seen in the figure below.

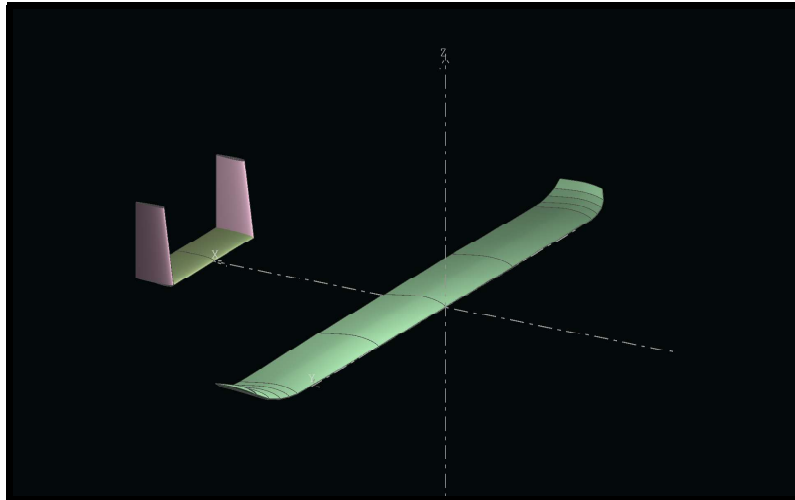


Figure 2. XFLR5 Initial Design

Table 1. Part Names and Dimensions

Part	Feature	Value (mm)
Wing	Root Chord	130
	Root Thickness	
	Tip Chord	90
	Tip Thickness	
	Wingspan	1300
Horizontal Stabilizer	Root and Tip Chord	80
	Root and Tip Thickness	
Vertical Stabilizer	Root Chord	80
	Root Thickness	
	Tip Chord	60
	Tip Thickness	

The Eppler 387 was chosen as the main wing's airfoil. The airfoil consists of a more convex upper half creating a gentle camber which generates more lift at lower speeds and lower angles of attack, conditions similar to the competition. XFLR5 Simulations were conducted in order to determine the parameters for the wing and elevator. In order to maximize aerodynamic efficiency, the wing has a mild sweep as well as winglets positioned at a 45 degree angle to reduce wing induced drag because of wing tip vortices. The H tail configuration was modified to have the vertical stabilizers sit slightly behind the leading edge of horizontal stabilizers. This configuration increased the aircraft's yaw stability. These configurations presented favourable results with a lift to drag ratio of 16.4 and a critical angle of 10.5 degrees.

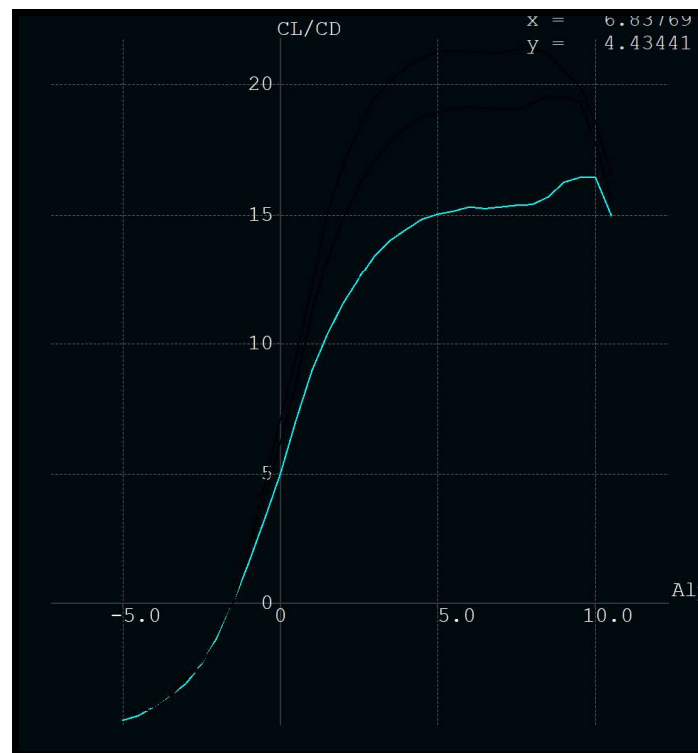


Figure 3. CL/CD vs Alpha graph

2.2 Aerodynamics

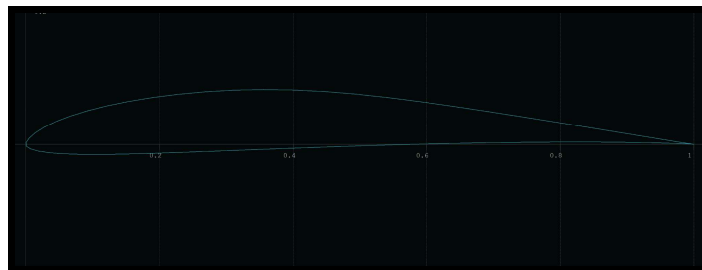


Figure 4. Eppler 387 Airfoil Shape

The Eppler 387 (E387) was chosen as the primary airfoil for the aircraft wing. Cambered airfoils are known for stronger lift performance and higher stall angles at low Reynolds numbers.

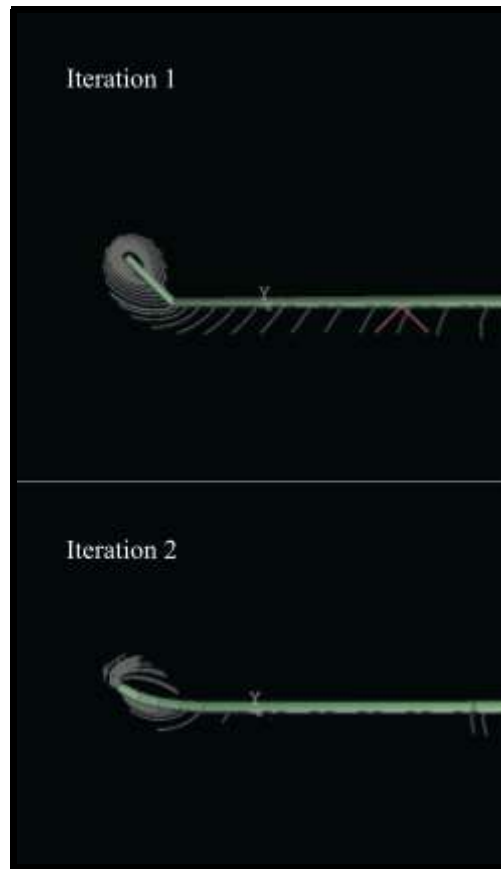


Figure 5. Wing tip vortices

In order to maximize lift to drag ratio, the wing features curved winglets designed to minimize wing tip vortices. The image above shows the difference between wing tip vortices for a standard winglet and a curved winglet, tested at the same wind speed and reynold numbers. The reduced induced drag provides significant benefits for both the gliding and the powered sections of the flight.

2.3 Propulsion/Power

With the creation of a sub 600 gram aircraft in mind, the team was able to use the smaller _ battery in conjunction with the smaller _ motor. The sizing of the battery with the exception of the motor was experimented with during iteration testing. This was done with two things in mind. Firstly, the aircraft requires a sufficient amount of speed in order to conform to the bounds of the football field while also being able to generate enough lift to stay in the air longer. For a lower sink rate, sufficient thrust would be required which is directly impacted by the power supplied by the battery.

Secondly, the size of the battery also shifted the Center of Gravity of the aircraft which impacted whether the aircraft was slightly nose-heavy or tail-heavy. Therefore different batteries were tested with, for example between the 4s and 3s batteries in order to accurately place the CG which ideally should be at quarter-chord. It was noticed that with the smaller sized Li-Po batteries, the aircraft would often be tail heavy and would noticeably perform poorly during flight. However with the heavier battery the center of gravity would be in a generally better position from before.

2.4 Systems and Electronics

The table below details the electronic components utilized in Vortex. All electronic components were picked based on the specifications of the battery and motor of the aircraft.

Table 2. Vortex Electronic Components

Component	Name	Mass (g)
Motor	DYS D2830 850kV / 1000 kV / 1300 kV Brushless Outrunner Motor 2-4S	88
ESC	Toytensi 50A ESC 2-4S	50
Battery	OVONIC 4S 1300mAh 14.8V LiPo Battery	150
Receiver	OrangeRx R615X 6Ch 2.4GHz Receiver	12
Servos	Smraza SG90 9G Micro Servo	9
Transmitter	Spektrum DX6e 6Ch Transmitter	-

2.5 Structures

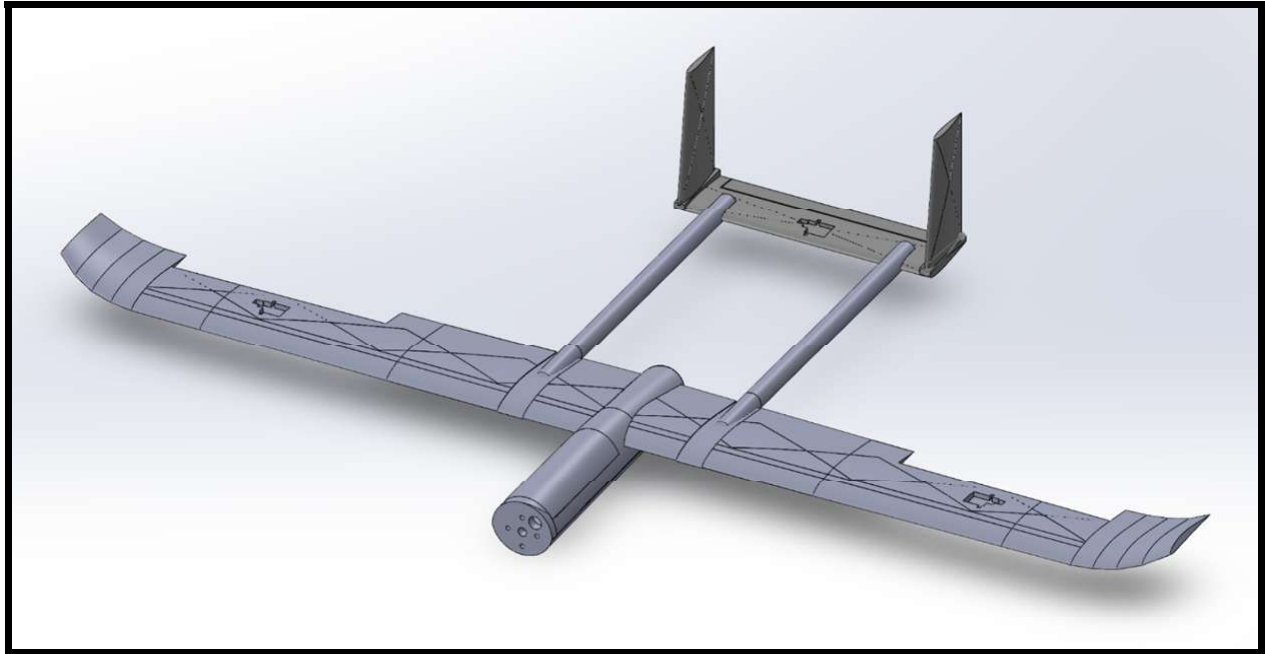


Figure 6. Vortex complete Solidworks design Isometric View

Vortex was 3D modeled in three key sections: the fuselage, the wing and twin boom, and the tail. The primary design driver for the structure was achieving a balance between lightweight design and structural integrity. The final design achieves this through a front-mounted motor configuration and specialized 3D printing techniques to minimize bending and overall weight.

Fuselage Design

The fuselage features a pull propeller configuration with the motor mount located at the front to ensure an ideal center of gravity. To reduce weight, the fuselage diameter was minimized, and the internal structure thickness was set to 0.4mm. The interior of the fuselage utilizes a triangular rib pattern spread around the rectangular electronics hatch to provide necessary rigidity while maintaining a lightweight profile.

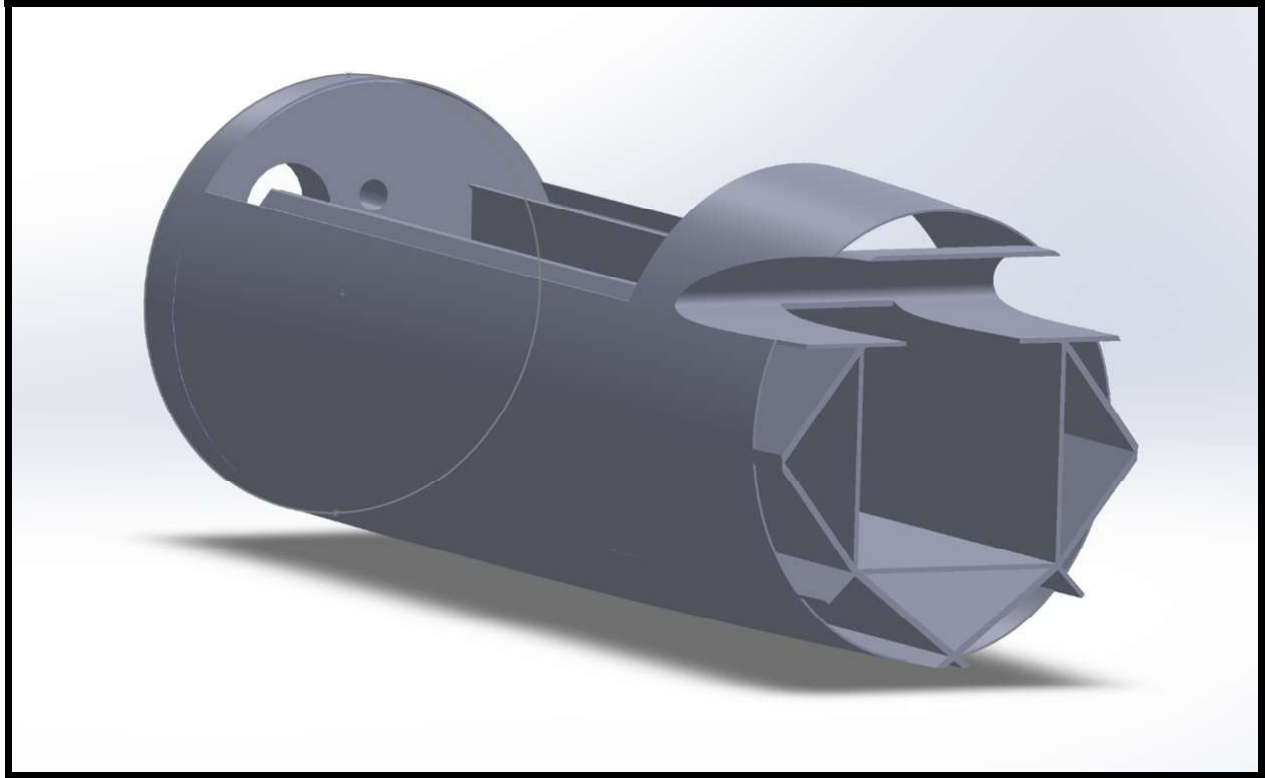


Figure 7. Section view of the internal ribs of the fuselage

Wing Design

The wing utilizes a specialized "Vase Mode" CAD technique, which enables the production of extremely thin and lightweight components. This method involves creating 0.1mm cuts through the wing body where internal structure is needed and then shelling out each individual piece made by these cuts. The gaps created by the cuts fuse together while printing creating a continuous internal structure with 0% infill. To maximize structural strength, the wing features integrated booms and a continuous spar. The connection points between the wing and booms were strengthened by printing them as single, non-vase mode pieces and a single continuous spar runs through the body of the wing to prevent fluttering and reduce the tendency to bend due to the weight of the booms.

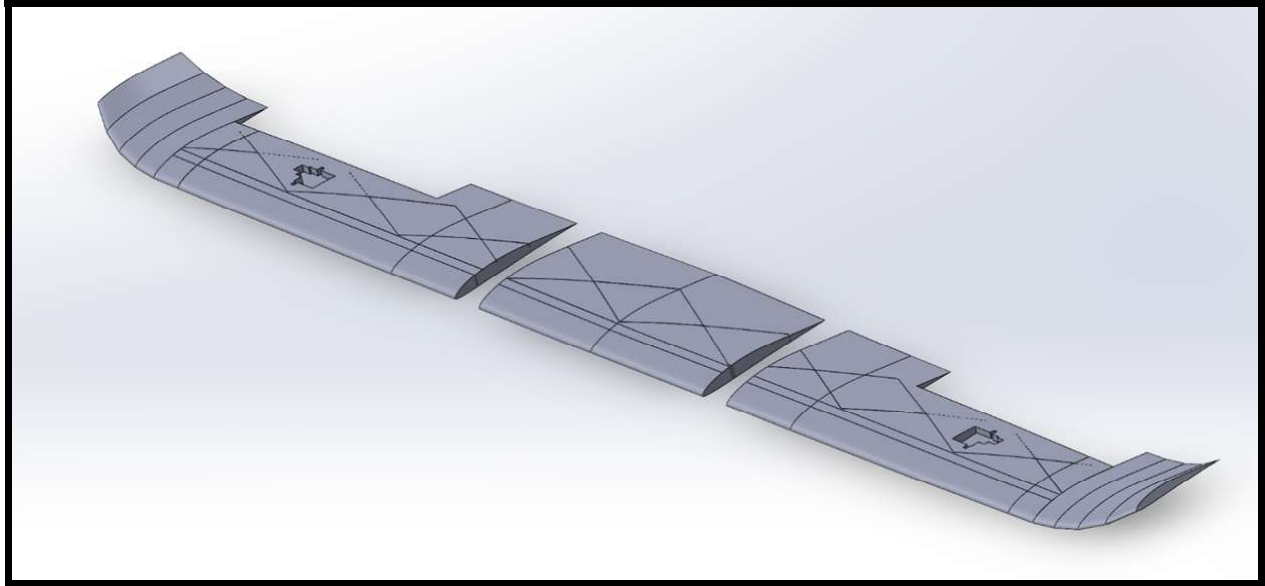


Figure 8. Vase mode sections of the wing

Booms and Tail Assembly

The design uses a twin-boom tail configuration with a boom spacing of 200mm. The distance between the booms was minimized to reduce bending moments on the wing. The booms have a 0.8mm wall thickness, no internal structure, and feature a flat bottom to facilitate easier assembly.

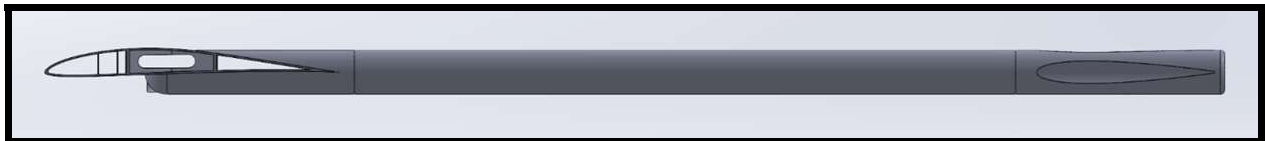


Figure 9. Integrated wings and boom side view

The tail assembly consists of horizontal and vertical stabilizers, both produced using the vase mode technique to minimize weight and feature an X internal structure. For a secure connection, the rear of the booms grips the leading edge half of the horizontal stabilizer, and implements a specialized connection piece that joins the vertical and horizontal stabilizers directly.

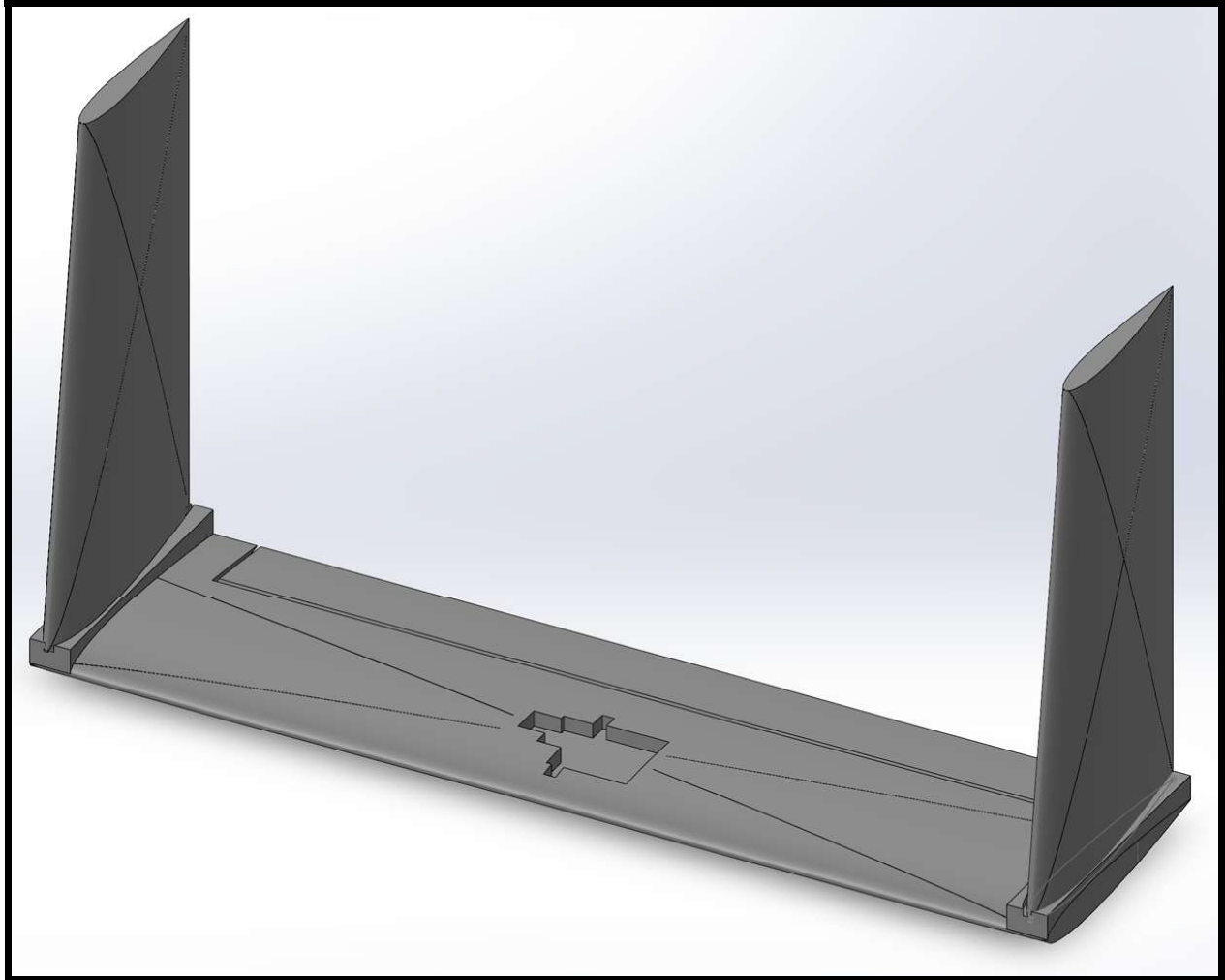


Figure 10. Tail Assembly Isometric View

3. Print and manufacturing

All aircraft components were produced using fused deposition modeling created on either a Bambu P1S 3D printer, an Ender 5 printer, a Creality K2 Plus 3D printer, or a Creality CR-30 print mill. The Bambu P1S and the Creality K2 Plus were used to create the aircraft body components, while the Creality CR-30 created supporting pieces such as box spars. All body components were printed with ColorFabb LW-PLA to reduce the weight of the parts significantly, while the spars were printed with MatterHackers PLA to increase rigidity. TPU hinges for the control surfaces were printed with MatterHackers TPU on the Ender 5 for increased flexibility and deflection.

The largest build height available among the printers used was 350 mm from the Creality K2 Plus, therefore all wing pieces were partitioned to maximize this dimension accordingly. However, the quality of the Creality K2 Plus parts was often unpredictable in comparison to the Bambu P1S parts. The Bambu P1S has a smaller build volume and would require the wing to be

partitioned into a greater number of sections. Therefore, careful design choices had to be made to balance quality with available volume, especially when printing with vase mode.

Vase mode is a slicer setting that allows the printer to create parts without retracting throughout the print. Deploying this feature significantly decreases print time and shell thickness, which in turn decreases weight by over 50% in comparison to non-vase parts. However, vase mode parts were often more difficult to print on the Creality K2 Plus and were more difficult to work with during manufacturing. In the Creality slicer, when vase mode is enabled, it automatically changes these settings: “enable supports”, “detect thin walls”, “wall loops”, and “sparse infill density”. Altering wall loops and sparse infill allows the printer to create uniform walls with no interior structure, but with no supports, printing the full 350 mm wing pieces became challenging. Many of the pieces would be printed with mismatched layer lines and a wall thickness significantly larger than what the slicer was set to.



Figure 11. Vase mode wing with layer lines

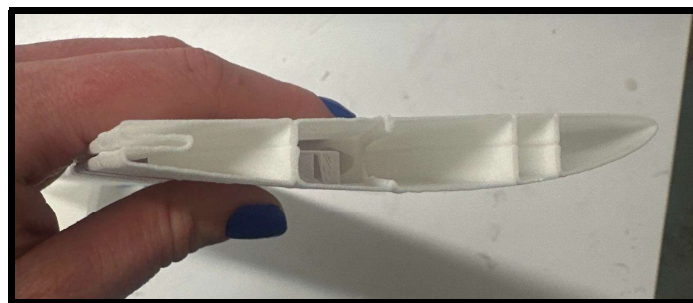


Figure 12. Thickness of vase mode wing

In order to produce more consistent parts, a series of experimental prints were created to optimize vase mode settings. Throughout these tests, flow rate, speed, outer wall thickness, layer height, and temperature were all tuned to produce parts with uniform layers and low weight. As a result, all of the vase mode wing pieces were produced with a nozzle temperature of 245 degrees Celsius, inner and outer wall speed of 25 mm/s, 0.24 mm layer height, and an outer wall thickness of 0.38 mm, all while being printed with a 0.40 mm nozzle.

Figure 13. Vase mode K2 Plus test prints

Vase mode also restricted the number of parts that could be printed at a single time. Since the printer cannot retract when printing an individual part in vase mode, the only way to put multiple parts on the same print bed is to enable “by object”, which reduces the maximum print height from 350 mm to 100 mm. This meant that parts more frequently had to be printed separately, and the net production time for parts, even with the lower individual print times, was often extended.

While there was a learning curve with the usage of these processes, applying them to the Bambu P1S and the Creality K2 Plus offered several advantages. The use of the Bambu P1S and Creality K2 Plus, especially with the addition of vase mode, significantly reduced the amount of post-processing that needed to be done on each part and increased the smoothness of the wing as a whole. Since the printer never retracts in vase mode, the parts were consistently more precise, allowing geometrically complex parts to be printed faster with less difficulty. Enabling “by object” was also particularly useful when printing multiple small body components, as there was a reduced likelihood of collisions and, consequently, fewer failed prints. As a result of these combined processes, the weight was significantly reduced from 750g to 590g, being the final weight. A sample part in the Bambu Studio slicer is shown in the figures below both with and

without vase mode enabled. Although there is a marginal difference in print time, the weight of this 180 mm part is cut in half with vase mode enabled.

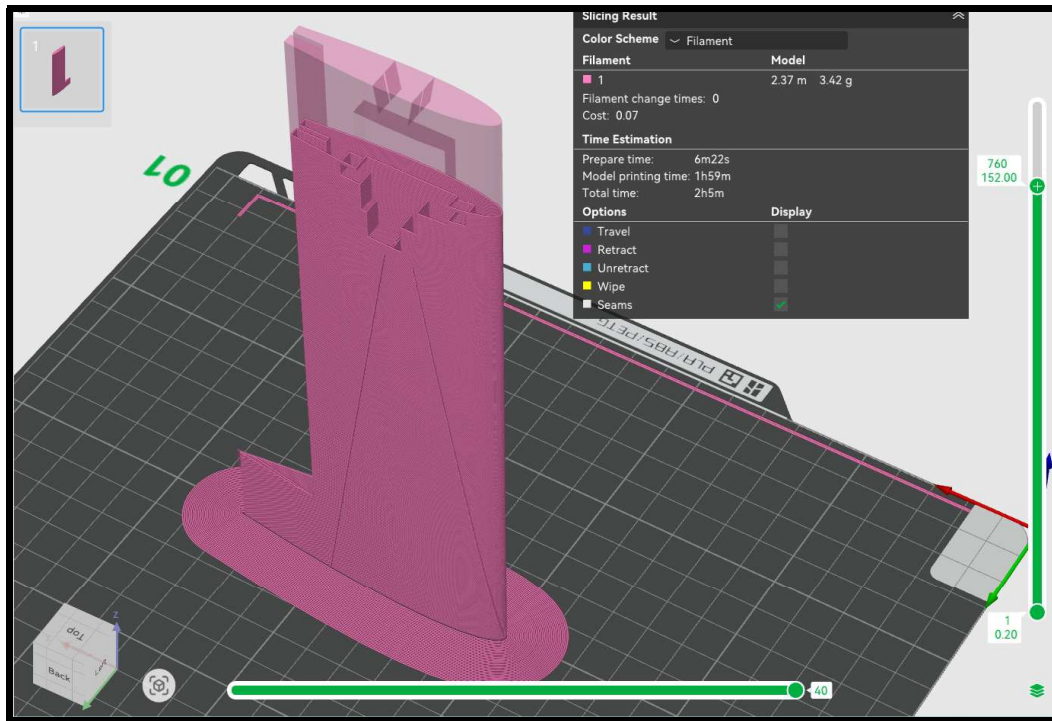


Figure 14. 180 mm part with vase mode enabled

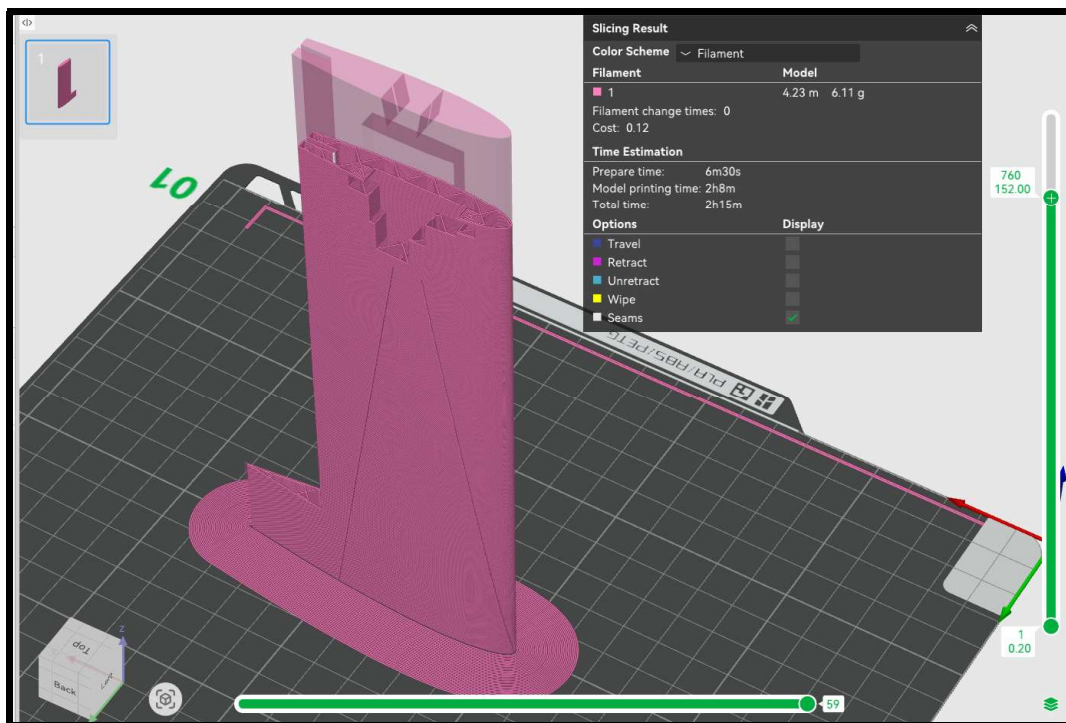


Figure 15. 180 mm part without vase mode

The biggest print challenge our team experienced, regardless of vase mode, was that some of the slicers had issues printing our internal structure and booms. The slicers would often not be able to identify the internal structure, especially in cases such as our wing pieces with the servo cutout. The Creality slicer would create large gaps in the parts that were not present in the CAD model, leading to a non-manifold geometry issue. This subsequently caused the internal structure to have to be recreated in CAD to allow the printer to account for the servo cutout.

The boom attachment to the wing was an especially difficult design challenge. Since the booms are integrated directly into the wing, there was often a significant amount of bending in the booms from the attachment, as seen in Figure 16.

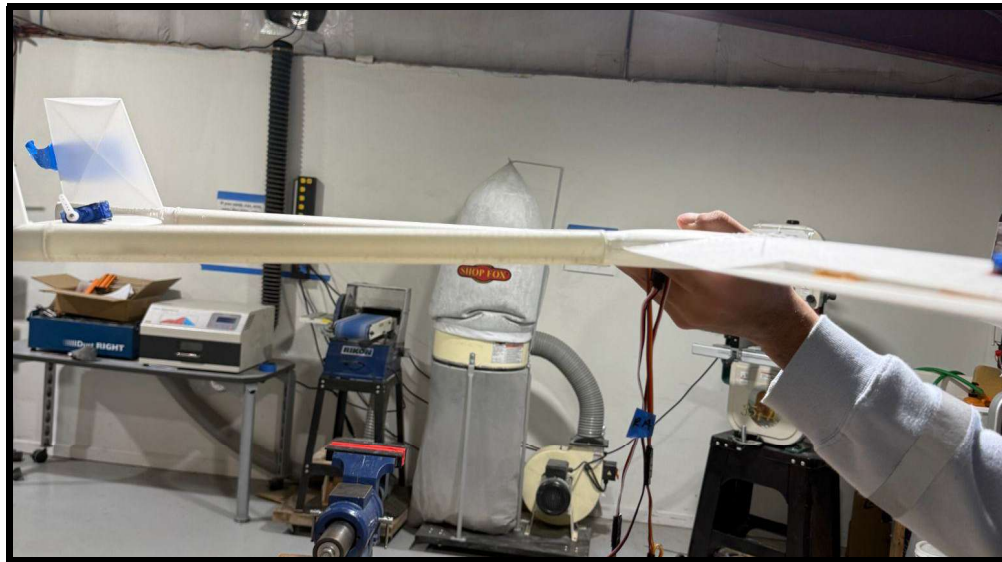


Figure 16. 180 mm part without vase mode

In an attempt to reduce the bending, the thickness of the booms was decreased, and male-to-female connectors were made to provide more support, but the issue persisted. The male-to-female connectors would often not print correctly because of the interference from supports. The few times that the connectors did print correctly, the tolerance on the booms made it difficult to properly put the connectors in place. The team considered printing the wing and boom pieces together as one continuous part, but the available print volume (350 mm x 350 mm) made this difficult. Printing the pieces combined also required the use of supports, which would be extremely difficult to remove once produced. Reluctantly, it was decided that the optimal way to simplify manufacturing while reducing the bend was to continue to split the boom at the wing and add triangular inserts to the top of the boom to strengthen the connection.

Another factor that may have contributed to the sag in the booms was the elevator wire. Our design required that the elevator servo wire run through one of the booms to be able to make the elevator operational. Although the wire weight was not significant enough on its own to cause the sag, it added tension along one side of the boom that was not present on the other side, contributing to the imbalance between the two.

Printing the middle boom pieces turned out to be quite a challenge in itself, even when split from the wing. When the full 343 mm boom piece is printed vertically with supports, it begins to wobble when the height surpasses 300 mm, and the booms would frequently fall over and subsequently fail. The team decided to integrate a flat face onto the booms so we could then print them horizontally. However, even with supports, there was consistently a large amount of overhang in the circular region of the boom, which led to the creation of several parts similar to the ones shown in Figure 17.



Figure 17. Overhang issues in the middle boom pieces

To combat this issue, the support threshold angle was increased such that supports would be added throughout the middle of the boom, and “bottom shell layers” were set to 3 so that the full bottom of the boom would be infilled. This did make removing the supports difficult during manufacturing, but the quality of the prints was much higher than when printed vertically.

Another considerable manufacturing issue with the design was keeping the vertical stabilizers straight. As proven by design analysis, it is crucial to the stability of the plane that the vertical stabilizers be perfectly vertical on the surface of the horizontal stabilizer. Originally, the booms had a connecting end piece that had an airfoil cutout for the vertical stabilizers to sit into. But when the inner boom distance was reduced, those pieces were no longer able to be used. The horizontal stabilizer has a curved top surface due to the shape of the airfoil, so attaching the vertical stabilizers directly with glue was a risky endeavor that required the constant use of a SCRIB3D 3D Printing Pen to fill large gaps. Resultantly, covers were developed to provide a slot for the vertical stabilizers to sit in without needing to be attached to the boom. The cover also proved to be difficult to work with. The Bambu Studio slicer software had difficulty reading the airfoil shape precisely, even with “print by object” enabled to print each cover independently. The covers were also very thin and did not leave a lot of clearance, so a significant amount of force had to be applied to put in the vertical stabilizers correctly.



Figure 18. Vertical stabilizer covers for the horizontal stabilizer

The assembled wing and boom configuration is shown in Figure 19 below.



Figure 19. Wing and Boom Assembly

4. Innovation, lessons

The Vortex aircraft was proven to utilize numerous innovative manufacturing techniques which were previously untested in the history of this team. The discovery of vase mode for the production of our aircraft introduced a whole new design concept which fundamentally changed how the team designed and manufactured aircraft. Vase mode enabled our wing to retain nearly the same structure and support while cutting down on weight significantly. With the pod-and-boom configuration however, vase mode was intensely more difficult to design for given that the aircraft's booms would not be able to be manufactured with the wing due to print support constraints. With this in mind, the team had to innovate again to create non-vase sections of the wing with boom inserts in order to account for this. The strength and support for the tail directly depended upon the stability of the joint connection between the non-vase and vase mode sections of the wing. The team, upon iterating this design many times, took many variables into account to come up with the ideal design for these sections, for example how far each boom would be placed from the center of the wing, non-vase wall thicknesses, booms' wall thicknesses, location of boom connection joint in terms of the chord length of the wing at that location, and many others. Accounting for every single variable to iterate upon in order to create a finalised ideal design required innovation in each iteration design cycle and manufacturing cycle.

With this in mind, a lot of lessons were learned along the way especially when it came down to striking a clean balance between structure, support and weight not just in the wing but also with the fuselage. Even with the fuselage, internal supports were cut down where not required in order to aid with a lower overall weight for the aircraft. Additionally, even with the discovery of vase mode which significantly reduced the weight of the aircraft, 0.1mm internal ribs were only kept in the middle of the wing with the skin of the wing being the only thing conforming to the airfoil shape in the leading and trailing edge. With this in mind, the density of the internals was also experimented with in order to make sure that weight was cut down where possible. Each of these innovative procedures and steps helped aid the creation of a sub 600 gram aircraft.

To conclude, combining vase-mode aircraft design along with a pod and boom configuration aircraft required a lot of innovative and unconventional thinking that helped facilitate the final iteration produced. With weight savings being the name of the game with fixed wing RC aircraft gliding, it was imperative that new and previously untested manufacturing and design processes be used. The team was able to adapt and overcome the obstacles in the way of weight savings with new procedures, which are exciting and also help pave the way for future aircraft design and production methods for the team,